Raw Examination Corpus Cohort

▶ HSC Science Stream Papers

n = 210 Papers (38.5% Cohort)

- Physics, Chemistry, Math, Biology
- 1st & 2nd Papers (Class 11–12 NCTB)
- 9 Boards | 2010–2024

▶ SSC Secondary Core Papers

n = 195 Papers (35.8% Cohort)

- General Science, Higher Math, Physics
- Grade 10 Standardized Benchmark
- Bloom's Cognitive Rubrics

▶ Commerce & Humanities Cohort

n = 140 Papers (25.7% Cohort)

- Accounting, Finance, Economics, Civics
- Non-STEM Cognitive Distribution
- Multi-Disciplinary Diversity

▶ Policy & Disruption Covariates

Curriculum Timeline C_cov

- 2020 COVID-19 Auto-Pass Hiatus
- 2021–2023 Shortened Syllabi
- Marks Shift (3/8 vs. 5/8 CQs)

Total Corpus: N = 545 Board Papers

15,307 Items | 56,149 Marks | 9 Boards

Multimodal Corpus Ingestion & Document Preprocessing

Dual-Stream Ingestion

Normalized Pixmap Renderer

- 150 DPI Bitonal Rasterization
- Contrast Histogram Stretching
- Margin Banner Filter
- Watermark Neutralizer

Bilingual OCR (ben+eng)

Tesseract Dual-Model Match

- Bengali script recognition
- English technical terms
- Math / equation symbols
- SHA-256 checksum

Regex Structural Parser

Cognitive Bloom Segmentation

- Stem (Uddipak) extraction
- 4 Sub-parts (a,b,c,d / Ka–Gha)
- MCQ 4-option match
- Checksum verification

[D_corpus]

Structured Question Corpus
Zero Data Leakage Buffer

- questions.jsonl (Structured CQs)
- questions.csv (15,307 rows)
- Temporal Graph G=(V,E,T)
- Train(≤17)|Val(18–22)|Test(23)

Hierarchical Educational Forecasting Engines

Chapter Prior Forecaster

Pillar 1: Temporal Graph Attention (TGAT)

$$\alpha_{uv}(t) \propto \exp(h_u \parallel h_v \parallel \Phi(t))$$

Marks-Weight Regressor

Pillar 2: Constrained Dirichlet Model

$$w_c \sim \mathrm{Dir}(\alpha), \quad \Sigma w_i = W_{total}$$

Curriculum Question Synthesizer

Pillar 3: Knowledge-Grounded RAG

$$P(S, q_a, q_b, q_c, q_d \mid T_{NCTB})$$

Cross-Board Diffusion Engine

Pillar 4: Inter-Division Migration

$$W(P_{Dhaka}, P_{Regional}) \le \tau$$

Calibrated Multi-Scale Decision Fusion & Loss Bounds

$L = \lambda_1 L_{TGAT} + \lambda_2 L_{Dir} + \lambda_3 L_{RAG} + \lambda_4 L_{Parity}$ | Syllabus Group Constraints

Recency / Out-of-Syllabus Alert

$P(c) < \varepsilon$ | Alternation Filter

Calibrated Assessment

High-Yield Chapter Prediction

HitRate@k ≥ 78% | Coverage ≥ 85%

Iterative Pedagogical Validation & Teacher Review Feedback Loop

Policy & Governance

▶ Policy & Recourse

- Non-Profit Academic Fair Use
- Public Ministry Archive Grounding
- Anti-Leak Diagnostic Framing
- Rural Education Parity Audit
- Double-Blind Teacher Review